# THE NOBLE GAS OF PRIMES

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Abstract. This note is expository. It develops a physical analogy for the collision program: in a fixed base, primes behave less like a reactive medium and more like a noble gas. The point is not that arithmetic is literally kinetic theory, nor that primes are particles in disguise. The point is structural. A local digit event, a residue landing in the same base-$b$ bin after multiplication, can be aggregated into a rigid, finite observable, the collision invariant. That observable survives the combinatorial chaos of long division in much the same way that a collision invariant in kinetic theory survives the microscopic chaos of a gas. The analogy clarifies why the words *collision* and *invariant* were chosen, and why the later transform and spectrum papers move so naturally from finite bin geometry to Dirichlet characters and $L$-functions.

## 1. Introduction

A prime is usually introduced negatively: it has no nontrivial divisors. That definition is complete, but it says little about how a prime behaves once a number base is fixed and long division is allowed to run. The collision program studies exactly that setting. For a fixed base $b$ and prime $p > b$, one partitions the nonzero residues $r \in \{1, \ldots, p-1\}$ into digit bins by the map

$$d_b(r) = \left\lfloor \frac{br}{p} \right\rfloor.$$

A *collision* occurs when $r$ and $gr \bmod p$ land in the same bin. The central discovery is that once these collisions are centered and aggregated, they produce rigid arithmetic structure: finite determination, reflection, harmonic decomposition, and eventually $L(1, \chi)$-encoding [1, 2, 3].

This note argues that the right physical analogy is not a reactive chemical medium but a noble gas. In a noble gas, the particles are stable, distinct, and nonbinding. They scatter, but they do not reorganize into complicated molecular composites. That is close to the role primes play in the collision program. Each prime carries its own digit geometry. The local events look irregular, but the aggregate obeys exact laws.

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The analogy should be read with care. It is not a claim that prime number theory reduces to physics. It is a claim that both subjects share a common mathematical shape: many local interactions, strong apparent randomness, and a surprisingly small family of quantities that survive averaging.

## 2. Why a Noble Gas

The word *noble* is useful here because it emphasizes stability without suggesting immobility. Noble gas atoms are not static. They move, collide, and distribute themselves throughout a container. What distinguishes them is that their interactions are clean. They do not typically lock themselves into larger bound structures. For kinetic theory, this makes them the ideal test case: one can study collisions without first having to model chemistry.

Primes play a similar role in the arithmetic setting. They are the stable, indivisible carriers of the digit geometry. At a prime modulus, multiplication by a unit permutes the nonzero residues cleanly across the $b$ bins. There is no Chinese remainder decomposition, no interleaving of separate prime factors, and no composite artifact created by the interaction of different local geometries. The prime case is therefore the purest setting in which to ask what long division preserves.

This is precisely why the composite case behaves differently. At composite moduli the same bin construction still exists, but the collision story becomes interleaved and partially factorized. The recent paper on partial character sums shows that the clean prime geometry gives way to a more complicated CRT mixture [4]. In that sense, primes are the noble gas of the theory: the medium in which the collision laws can be seen before reactions begin.

## 3. The Microscopic Event

The microscopic event is simple. Choose a base $b$, a prime $p > b$, and a unit multiplier $g \in (\mathbb{Z}/p\mathbb{Z})^\times$. Then compare the bins of $r$ and $gr$. The collision count is

$$C_p(g) = \#\Big\{r \in \{1, \dots, p-1\} : d_b(r) = d_b(gr \bmod p)\Big\}.$$

Nothing in this definition looks analytic. It is an integer count attached to a finite permutation of residues.

But the point of the theory is that these local coincidences are not the end of the story. One centers them,

$$S_p(g) = C_p(g) - \frac{p-1}{b},$$

and then studies the resulting deviation function rather than the raw count. The collision invariant is the part of this data that remains rigid under the chaotic-looking rearrangement of residues.

That is where the kinetic analogy begins to earn its keep. In a gas, a single collision is unimportant. What matters is the structured quantity that remains visible after one averages over many collisions. In the prime setting, a single bin coincidence is likewise too small to carry meaning on its own. The meaning appears only after the coincidences have been aggregated into a finite law.

## 4. The Conserved Coarse Data

In kinetic theory, a collision invariant is not merely a quantity that happens to agree before and after one encounter. It is the coarse data that survives the entire microscopic churn of the system. It is what remains available for the macroscopic equations.

The arithmetic analogue is not literal conservation of mass or momentum. It is finitary persistence. The collision deviation is not destroyed by the internal rearrangement of residues. Instead, it collapses to a finite arithmetic stamp. The finite determination theorem shows that at lag $\ell$, the quantity $S_\ell(p)$ depends only on the residue class of $p$ modulo $b^{\ell+1}$ [1]. That is one of the decisive structural moments of the theory. A phenomenon that first appears to depend on the full prime is already determined by a bounded local datum.

This is why the word *invariant* feels historically loaded rather than accidental. The object is not invariant because nothing moves. It is invariant because something global survives movement. The residues are permuted, the digit positions vary, and the local collisions fluctuate, but the centered aggregate remains governed by a finite law.

## 5. Reflection as Reciprocity

The reflection identity is one of the clearest places where the analogy becomes conceptually strong. In the collision invariant paper, complementary classes are not free to vary independently. They satisfy an exact balancing law [1]:

$$S_\ell(a) + S_\ell(m - a) = -1.$$

This is not conservation in the Newtonian sense. No physical quantity is being transported from one side of the equation to the other. But it is very close in spirit to a conservation law because it expresses an exact reciprocity imposed by the underlying geometry.

In that sense, the reflection identity plays the role of an arithmetic balance law. Once one side is known, the other side is determined. The observable is not free to drift. That is precisely the kind of rigidity that makes the gas analogy useful. The system contains local motion but not arbitrary large-scale freedom.

One can also describe this as a law for arithmetic echoes. The values $S_\ell(a)$ and $S_\ell(m-a)$ are mirror responses of the same finite geometry. A collision excess on one side cannot appear without a compensating deficit on the other, because the pair is locked to an exact sum. That is the right sense in which the reflection identity resembles a conservation law: not because a physical substance is being preserved, but because asymmetry is not free to accumulate across the reflected classes.

It is also the point at which the metaphor should be handled carefully. One does not want to say that the reflection identity *is* momentum conservation. That would be false. What one can say is that both statements share the same formal attitude: a many-body looking system is secretly constrained by a low- dimensional exact law.

## 6. From Local Collisions to Harmonic Modes

The transform paper pushes the analogy beyond static counting. In physics one often passes from individual collisions to spectral variables, normal modes, or hydrodynamic quantities. The arithmetic story does something similar. The collision deviation is decomposed against Dirichlet characters, and the resulting transform isolates the odd-character sector [2]. In that sense the reflection law already acts like a filter: the mirrored balance removes the even contribution and forces the surviving harmonic content into an odd arithmetic channel. The collision spectrum then shows that the Fourier coefficients factor into explicit arithmetic pieces and encode $L(1,\chi)$-values [3].

This matters conceptually. The local event is bin-level and combinatorial. The spectral outcome is harmonic and analytic. The bridge between them is the real content of the program. Once that bridge exists, the collision invariant no longer looks like a curious statistic extracted from long division. It becomes a finite interface between digit geometry and the analytic structure of $L$-functions.

That is another reason the noble-gas analogy is useful. One does not study a gas only by watching particles bounce. One studies the emergent collective variables that remain meaningful at scale. The collision transform is exactly that kind of move. It passes from local encounters to global modes.

## 7. Pressure, Flow, and the Prime Medium

The analogy becomes most tempting when one begins to talk about prime distribution itself. A gas exerts pressure, reaches equilibrium, and obeys transport laws not because individual particles move predictably, but because the aggregate has rigid statistical structure. The primes exhibit a comparable tension: individually they appear scattered, yet collectively they obey strong regularities and are governed by the zeros of zeta and Dirichlet $L$-functions.

The collision program does not prove that primes are a gas, nor does it replace analytic number theory with fluid mechanics. But it does encourage a new perspective. Long division in base $b$ creates a medium in which primes interact with digit bins, and the resulting collision statistics produce a finite, harmonic, law-governed structure. If one wants a physical image, the right image is not a crystal and not a chemical reactor. It is a sparse, stable gas whose microscopic events are individually simple but whose coarse behavior is highly constrained.

This language is especially useful when discussing why the later papers turn so quickly toward strip questions, cancellation, neutrality, polarity, and zero- density phenomena [5, 6]. Once one has a transform into character space, the natural questions are no longer purely combinatorial. They become questions about flow, balance, and collective interference across the harmonic modes.

## 8. Where the Analogy Stops

The analogy must stop in several places.

First, the theory does not redefine primality. A prime remains a prime in the classical sense. The collision program adds structure *on top of* that classical definition once a base is fixed.

Second, the collision invariant is not literally a conserved physical quantity. No energy or momentum is being transported through arithmetic space. The correct claim is weaker and more precise: the centered collision observable is rigid, finite, and transformable in a way that parallels the role of collision invariants in kinetic theory.

Third, the prime case is special precisely because it is clean. At composite moduli the story becomes more reactive. Partial character sums over bins still exist, but the geometry is no longer that of a single stable atomic species. This is why the noble-gas analogy belongs first to the prime setting.

Finally, nothing in this note should be read as a proof strategy for the Riemann Hypothesis or the Generalized Riemann Hypothesis. The analogy is explanatory, not conclusive. Its value lies in clarifying what

kind of object the collision invariant is: not a random counting trick, but a stable arithmetic remnant of a microscopic digit process.

## 9. A Spectral Test

The noble gas analogy makes a prediction that can be checked. In kinetic theory, the collision invariants of a gas are the quantities that survive thermalization. They occupy the low-energy sector of the spectrum. The thermal modes carry the bulk statistics. The invariants carry the structured residue.

If the analogy is structural and not merely verbal, then the collision table should populate the low-energy spectral modes when the primes are given a coherence geometry, while the bulk spectrum should thermalize to a universal profile.

Watson [7] provides the framework for this test. He constructs self-adjoint Hamiltonians on the first $N$ primes using pairwise divergences and extracts spectral observables: heat trace, entropy, and eigenvalue growth. His central finding is spectral universality. Across logarithmic, entropic, and power-law divergences, the resulting spectral dimension $d_s \approx 1/2$ is the same. The bulk spectrum of the primes is insensitive to how one measures distance between them.

The collision table supplies a divergence that Watson's framework does not consider. For a fixed base $b$, set

$$\delta_{ij}^{(\mathrm{coll})} = |S(p_i \bmod b^2) - S(p_j \bmod b^2)|,$$

where $S$ is the collision invariant at lag $\ell = 1$. This is base-dependent by construction and takes finitely many values.

When this divergence is fed into Watson's coherence Hamiltonian and the eigenvalues are computed for $N = 300$ primes in bases 7, 10, and 12, two things happen.

First, the bulk spectral profile is unchanged. The heat trace and spectral dimension match Watson's base-independent results in every base tested. The large-scale spectral compression of the primes is reproduced.

Second, the low-energy spectrum separates. The first nontrivial eigenvalue $\lambda_2$ is base-dependent:

| Base | $\lambda_2^{(\mathrm{coll})}$ | $\lambda_2^{(\mathrm{log})}$ |
|---|---|---|
| 7 | 0.570 | 1.000 |
| 10 | 0.670 | 1.000 |
| 12 | 0.701 | 1.000 |

The logarithmic divergence places $\lambda_2 = 1$, coinciding with the bulk at this kernel scale. The collision divergence resolves additional structure: a separated, base-dependent mode. The first $\sim$10 eigenvalues are non-degenerate and vary with the base. Beyond the tenth eigenvalue the spectrum saturates. (The construction, a $\delta_0$ sweep confirming robustness, and spectral profiles are given in Appendix A.)

That is the spectral realization of the gas analogy. The bulk thermalizes: its profile is universal. The collision invariant populates the surviving low-energy modes: its structure is base-dependent. The modes that resist thermalization are exactly the ones the collision table sees.

## 10. Conclusion

The phrase *collision invariant* was a strong choice of name because it carries a strong conceptual promise. It suggests that long division, despite its apparent combinatorial noise, contains quantities that survive averaging and therefore deserve to be treated as law-like.

The prime case is where this promise is easiest to see. Primes behave like the noble gas of the theory: stable, noncomposite, and clean enough for the collision laws to emerge without interference from factorization. Their local bin collisions are the microscopic events. The centered collision deviation is the coarse invariant. Reflection supplies reciprocity. The transform produces harmonic modes. The spectrum reaches $L$-values. And when the collision table is placed inside a spectral geometry, it populates the low-energy modes that resist thermalization while the bulk spectrum universalizes. The surviving part is base-dependent. The thermal part is not. That is the spectral signature of a collision invariant.

Taken together, these features do not turn arithmetic into physics. They do something subtler and more useful. They show that the arithmetic of primes in a fixed base admits the same kind of question that kinetic theory asks of a gas: what survives the local chaos, and what laws govern the surviving part? The collision program is compelling because it offers a concrete answer. The spectral test is compelling because it confirms that the answer has the right shape.

## Appendix A. Spectral Computation

This appendix records the details of the spectral test described in Section 8.

A.1. **Setup.** Let $\mathbb{P}_N = \{p_1 < \cdots < p_N\}$ be the first $N$ primes. For each base $b$ and prime $p_i$, the collision invariant value is $S_i = S(p_i \bmod b^2)$

from the collision table at lag $\ell = 1$. The collision divergence is $\delta_{ij} = |S_i - S_j|$.

Following Watson [7], the unnormalized coherence kernel is

$$K_{ij} = \exp\Big(-\frac{\delta_{ij}}{\delta_0}\Big),$$

where $\delta_0 = (\max S - \min S)/3$ for the collision divergence and $\delta_0 = 2\log p_N/3$ for Watson's logarithmic divergence $\delta_{ij}^{(\log)} = \log p_i + \log p_j$.

The degree matrix $D = \mathrm{diag}(\sum_j K_{ij})$, the symmetric normalization $\widetilde{K} = D^{-1/2} K\, D^{-1/2}$, and the normalized Laplacian $L = I - \widetilde{K}$ yield the order-4 coherence Hamiltonian $H = L^2$. Eigenvalues are computed by LAPACK (`dsyev`).

A.2. **Heat trace profiles.** The heat trace $\Theta(t) = \sum_k e^{-t\lambda_k}$ and the spectral dimension $d_s(t) = -2\, d\log\Theta/d\log t$ were computed for $N = 300$ primes in bases $b = 7, 10, 12$ and for Watson's logarithmic divergence.

All four profiles agree within plotting resolution. The spectral dimension rises from near zero at small $t$, reaches a peak $d_s \approx 6.3$ near $t \approx 6$, and decays rapidly toward zero at large $t$. No configuration produces a region of constant $d_s$, confirming Watson's finding that the prime coherence geometry is sub-Euclidean and does not stabilize.

A.3. **Eigenvalue structure.** For each base, the collision Hamiltonian has ∼10 non-degenerate eigenvalues below the bulk saturation at $\lambda = 1$. The number of resolved low-energy modes is consistent with the number of distinct $S$-values in the collision table: 12 in base 7 ($\varphi(49) = 42$ coprime classes), 17 in base 10 ($\varphi(100) = 40$ classes), and 22 in base 12 ($\varphi(144) = 48$ classes).

The first five eigenvalues of the collision Hamiltonian:

| $k$ | Base 7 | Base 10 | Base 12 |
|---|---|---|---|
| 1 | 0.000 | 0.000 | 0.000 |
| 2 | 0.570 | 0.670 | 0.701 |
| 3 | 0.725 | 0.748 | 0.784 |
| 4 | 0.805 | 0.832 | 0.857 |
| 5 | 0.870 | 0.902 | 0.896 |

For comparison, the logarithmic divergence produces $\lambda_k = 1.000$ for all $k \geq 2$ at this kernel scale.

A.4. **Robustness under $\delta_0$ variation.** The coherence length $\delta_0$ controls the decay rate of the kernel. A natural concern is whether the separation of $\lambda_2$ in the collision case is an artifact of the particular $\delta_0$

chosen. To test this, we sweep $\delta_0$ from $0.1\times$ to $2.0\times$ the natural range of each divergence (the $S$-value range for the collision divergence, $2\log p_N$ for the logarithmic divergence) and report $\lambda_2$ in each case (base 10, $N = 300$):

| $\delta_0$ scale | $\lambda_2^{(\mathrm{coll})}$ | $\lambda_2^{(\mathrm{log})}$ |
|---|---|---|
| $0.1\times$ | 0.062 | 1.000 |
| $0.3\times$ | 0.636 | 1.000 |
| $0.5\times$ | 0.771 | 1.000 |
| $1.0\times$ | 0.878 | 1.000 |
| $1.5\times$ | 0.917 | 1.000 |
| $2.0\times$ | 0.937 | 1.000 |

The collision divergence resolves a separated $\lambda_2$ at every value of $\delta_0$ tested. The logarithmic divergence does not resolve $\lambda_2$ at any value. The fine structure is not a scaling artifact. It is a consequence of the finite, discrete collision table imposing block structure on the kernel.

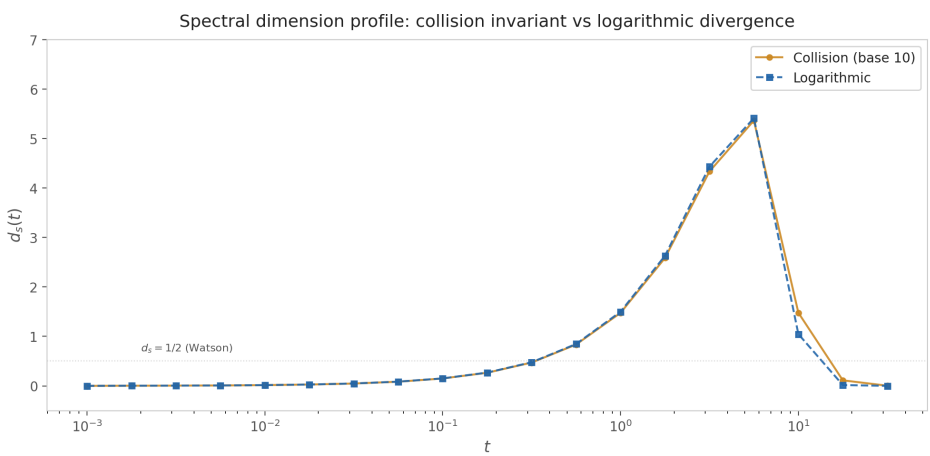


FIGURE 1. Spectral dimension profile $d_s(t)$ for the collision divergence (base 10, gold) and the logarithmic divergence (blue). The profiles are indistinguishable at $N = 300$, confirming bulk universality.

A.5. **Remarks.** The computation was performed at $N = 300$. Extending to higher lag $\ell$ in the collision invariant would produce finer divergences with $\varphi(b^{\ell+1})$ classes and potentially additional resolved spectral modes.

Source code: `experiments/spectral_collision_lapack.c` in the `nfield` repository.

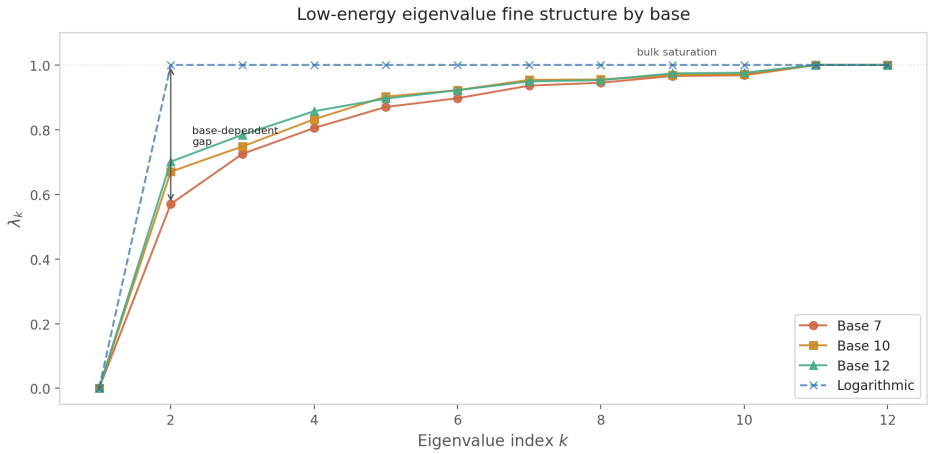


FIGURE 2. Low-energy eigenvalues of the coherence Hamiltonian by base. The collision divergence (colored curves) resolves base-dependent fine structure below the bulk saturation at $\lambda = 1$. The logarithmic divergence (dashed) coincides with the bulk at this kernel scale.

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